

Topic B1

Characteristics and classification of living organisms

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B1.1	Characteristics of living organisms
B1.2	Concept and use of classification systems
B1.3	Features of organisms

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Characteristics of life

Living organisms show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. These characteristics describe coordinated cellular processes; no single characteristic alone is always sufficient to decide whether an unfamiliar object is living.

Examiner insight

Distinguish respiration, a cellular chemical process, from breathing.

2. Classification

Classification places organisms into groups using shared features. Modern systems also use DNA base-sequence evidence to infer evolutionary relationships. Binomial naming gives each species a two-part Latin name: genus followed by species.

Examiner insight

A genus name starts with a capital letter and the species name with lower case; both are italicised when typed.

3. Keys and major groups

Dichotomous keys use paired contrasting statements to identify organisms. Vertebrates are separated using features such as body covering, reproduction and temperature control. Arthropods have segmented bodies, jointed legs and an exoskeleton; their classes differ in legs, antennae and body regions.

Examiner insight

Use visible, mutually exclusive features in each pair of a key.

4. Microorganisms

Plants have cellulose cell walls and chloroplasts; animals lack cell walls and chloroplasts; fungi have chitin walls and feed by extracellular digestion; bacteria are prokaryotic; viruses contain genetic material inside a protein coat and reproduce only inside host cells.

Examiner insight

Viruses are not cells and do not carry out independent metabolism.

Worked application

Situation: A newly found organism has chitin cell walls and absorbs digested nutrients.

Prediction: Classify it as a fungus.

Reason: Chitin walls and absorptive nutrition are characteristic of fungi.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B1

Characteristics and classification of living organisms

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines species? [1]
 - A a group of organisms that can reproduce to produce fertile offspring
 - B a two-part scientific name consisting of genus and species
 - C a sequence of paired contrasting statements used for identification
 - D an animal with a vertebral column
2. Which statement best defines binomial system? [1]
 - A a sequence of paired contrasting statements used for identification
 - B a two-part scientific name consisting of genus and species
 - C an animal with a vertebral column
 - D genetic material enclosed in a protein coat that replicates only in host cells
3. Which statement best defines dichotomous key? [1]
 - A an animal with a vertebral column
 - B genetic material enclosed in a protein coat that replicates only in host cells
 - C a sequence of paired contrasting statements used for identification
 - D a group of organisms that can reproduce to produce fertile offspring
4. Which statement best defines vertebrate? [1]
 - A genetic material enclosed in a protein coat that replicates only in host cells
 - B a group of organisms that can reproduce to produce fertile offspring
 - C a two-part scientific name consisting of genus and species
 - D an animal with a vertebral column
5. Which statement best defines virus? [1]
 - A genetic material enclosed in a protein coat that replicates only in host cells
 - B a group of organisms that can reproduce to produce fertile offspring
 - C a two-part scientific name consisting of genus and species
 - D a sequence of paired contrasting statements used for identification
6. Which statement about characteristics of life is correct? [1]
 - A Organisms that merely look similar.
 - B Living organisms show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.
 - C A common name in two languages.
 - D A food web with two branches.
7. Which statement about classification is correct? [1]
 - A A common name in two languages.
 - B A food web with two branches.
 - C Classification places organisms into groups using shared features.
 - D Any animal with an exoskeleton.

8. Which statement about keys and major groups is correct? [1]
- A A food web with two branches.
 - B Any animal with an exoskeleton.
 - C A small prokaryotic cell.
 - D Dichotomous keys use paired contrasting statements to identify organisms.
9. Which statement about microorganisms is correct? [1]
- A Plants have cellulose cell walls and chloroplasts; animals lack cell walls and chloroplasts; fungi have chitin walls and feed by extracellular digestion; bacteria are prokaryotic; viruses contain genetic material inside a protein coat and reproduce only inside host cells.
 - B Any animal with an exoskeleton.
 - C A small prokaryotic cell.
 - D Organisms that merely look similar.
10. Which prediction is most appropriate when a newly found organism has chitin cell walls and absorbs digested nutrients? [1]
- A Organisms that merely look similar.
 - B A food web with two branches.
 - C Classify it as a fungus.
 - D A small prokaryotic cell.

Section B · Structured questions

11 Biologists use precise vocabulary when studying characteristics and classification of living organisms.

(a) Define species. [2]

(b) Define binomial system. [2]

(c) Distinguish dichotomous key from the incorrect idea: “a food web with two branches”. [2]

(d) State two important features of characteristics of life. [2]

[Total: 8]

12 A newly found organism has chitin cell walls and absorbs digested nutrients.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records matches with reference species under three conditions.

condition	matches with reference species
species A	92
species B	71
species C	48

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to use visible features to construct a dichotomous key for four leaves.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of characteristics and classification of living organisms to an unfamiliar context.

(a) Define vertebrate. [2]

(b) Explain the biological importance of keys and major groups. [3]

(c) Correct this statement: “a small prokaryotic cell”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A group of organisms that can reproduce to produce fertile offspring.
- 2 B. A two-part scientific name consisting of genus and species.
- 3 C. A sequence of paired contrasting statements used for identification.
- 4 D. An animal with a vertebral column.
- 5 A. Genetic material enclosed in a protein coat that replicates only in host cells.
- 6 B. Distinguish respiration, a cellular chemical process, from breathing.
- 7 C. A genus name starts with a capital letter and the species name with lower case; both are italicised when typed.
- 8 D. Use visible, mutually exclusive features in each pair of a key.
- 9 A. Viruses are not cells and do not carry out independent metabolism.
- 10 C. Chitin walls and absorptive nutrition are characteristic of fungi.

Solutions · Structured questions

11. (a) a group of organisms that can reproduce to produce fertile offspring [2]. (b) a two-part scientific name consisting of genus and species [2]. (c) a sequence of paired contrasting statements used for identification; not a food web with two branches [2]. (d) Credit any two accurate features from the guide: Living organisms show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. These characteristics describe coordinated cellular processes; no single characteristic alone is always sufficient to decide whether an unfamiliar object is living. [2].

12. (a) Classify it as a fungus [2]. (b) Chitin walls and absorptive nutrition are characteristic of fungi [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $92 - 48 = 44$ [2]. (b) The values change from 92 to 48, with the highest value 92. [2]. (c) More matching bases indicate a closer evolutionary relationship [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Leaf shape or margin [1]. (b) Correct identification [1]. (c) Same maturity and undamaged specimens [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Test the key with an independent user [1].

15. (a) an animal with a vertebral column [2]. (b) Award linked points from: Dichotomous keys use paired contrasting statements to identify organisms. Vertebrates are separated using features such as body covering, reproduction and temperature control. Arthropods have segmented bodies, jointed legs and an exoskeleton; their classes differ in legs, antennae and body regions. [3]. (c) Correct statement: virus means genetic material enclosed in a protein coat that replicates only in host cells [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.